Response to the final report

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Seven-CNOT free-angle construction. Fine-step continuation with a fixed local gauge produced a word whose only varying local angles are $\pi - 2\theta$ and $2\pi - 2\theta$. Rational-angle reconstruction and full symbolic matrix multiplication verify the literal identity for every real θ, including the negative-reflection variant. It uses seven CNOTs and eighteen physical U_3 gates, with no GP gate or residual scalar. The complete time-ordered table is in the supplement; the exact JSON word and regression checks are archived. The proved interior bounds are now $4 \leq E_{\mathrm{CX}} \leq 7$ and $3 \leq E_{\mathrm{arb}} \leq 5$. Five/seven minimality remains a conjecture.

Explicit obstruction scope. The supplement now names the alternating time-ordered edge words (ca, ab, ca, ab), (ab, ca, ab, ca), (cb, ab, cb, ab), and (ab, cb, ab, cb), under the condition that the two control-edge gates are individually controlled on c in its computational basis. The zero-branch normalization then forces the prohibited doubled spectrum. The branch equation is written out. No normalization of arbitrary control-edge gates is assumed, and other four-gate layouts remain outside this partial argument.

Certificates and numerical fits. The supplement uses the two-SYC B certificate to explain why an unsuccessful local-frame fit cannot disprove exact reachability. It does not claim that we reproduced the referee's 64-restart experiment. Count certificates and explicit shortest symbolic local frames remain separate deliverables.

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